

Dayflow HRMS

API Contract Specification

Backend–Frontend Integration Document

1 Purpose of this Document

This document defines the complete API contract for the Dayflow Human Resource Management System (HRMS). It serves as the single source of truth for backend and frontend development.

All data exchange between frontend and backend **MUST** follow this specification exactly.

2 System Overview

- Architecture: API-first
- Backend: FastAPI
- Frontend: React
- Database: SQLite (MVP)
- Data format: JSON
- Authentication: JWT (Bearer Token)

3 User Roles

- **EMPLOYEE**: Can access only self-related data
- **ADMIN**: HR/Admin with full management privileges

4 Authentication and User Creation Rules

- Public user signup is **not allowed**
- The initial Admin user is seeded by the backend
- Only Admin users can create employee accounts
- Employees authenticate using a system-generated **Login ID**
- Login IDs are immutable
- Initial employee passwords are auto-generated by the system
- Employees may change their password after login

4.1 Login ID Generation

Login IDs are generated using the following format:

OI + first two letters of first name + first two letters of last name
+ year of joining + serial number for that year

Example:

OIJOJO20220001

5 Authentication Rules

- All APIs except /auth/login require authentication
- JWT token must be passed in the HTTP header

Authorization: Bearer <JWT_TOKEN>

6 API Summary

Feature	Endpoint	Role
Login	POST /auth/login	All
Get Current User	GET /auth/me	All
Create Employee	POST /admin/users	Admin
View All Employees	GET /admin/users	Admin
View Own Profile	GET /users/me	Employee
Update Own Profile	PUT /users/me	Employee
Change Password	PUT /users/me/password	All
Check-in Attendance	POST /attendance/check-in	Employee
Check-out Attendance	POST /attendance/check-out	Employee
View Own Attendance	GET /attendance/me	Employee
View All Attendance	GET /admin/attendance	Admin
Apply Leave	POST /leave/apply	Employee
View Own Leave Requests	GET /leave/me	Employee
View All Leave Requests	GET /admin/leave	Admin
Approve/Reject Leave	PUT /admin/leave/id	Admin
View Own Payroll	GET /payroll/me	Employee

7 Detailed API Specifications

7.1 Authentication APIs

7.1.1 POST /auth/login

Authenticate user using Login ID.

Request Body:

```
{
  "login_id": "OIJOJO20220001",
  "password": "password123"
}
```

Response:

```
{
  "access_token": "jwt-token-string",
  "role": "EMPLOYEE"
}
```

7.1.2 GET /auth/me

Fetch authenticated user details.

```
{
  "id": 1,
  "login_id": "OIJOJO20220001",
  "name": "John Doe",
  "role": "EMPLOYEE"
}
```

8 Admin Employee Management

8.0.1 POST /admin/users

Create a new employee account.

Request Body:

```
{
  "first_name": "John",
  "last_name": "Doe",
  "year_of_joining": 2022,
  "email": "john@company.com",
  "phone": "9999999999"
}
```

Response:

```
{
  "login_id": "OIJOJO20220001",
  "temporary_password": "AutoGen@123"
}
```

9 Employee Profile

9.0.1 GET /users/me

```
{
```

```
"login_id": "OIJOJO20220001",
"name": "John Doe",
"email": "john@company.com",
"phone": "9999999999",
"job_title": "Software Engineer"
}
```

9.0.2 PUT /users/me

```
{
  "phone": "8888888888",
  "address": "Surat"
}
```

9.0.3 PUT /users/me/password

Change account password.

```
{
  "old_password": "AutoGen@123",
  "new_password": "NewStrongPass@1"
}
```

10 Attendance Management

Attendance status is derived by the backend:

- Approved leave → ON_LEAVE
- Attendance record exists → PRESENT
- Otherwise → ABSENT

11 Payroll

Payroll information is read-only. Salary computation based on attendance is **out of scope for MVP**.

12 Feature Call Order

12.1 Employee

1. POST /auth/login
2. GET /auth/me
3. GET /users/me
4. POST /attendance/check-in
5. POST /leave/apply

12.2 Admin

1. POST /auth/login
2. POST /admin/users
3. GET /admin/users
4. GET /admin/attendance
5. GET /admin/leave

13 Final Notes

- Frontend must not enforce security rules
- Backend is the sole authority for access control
- All admin endpoints are protected under /admin/ *